

Nonlinear Shrinkage Estimation of Large Integrated Covariance Matrix

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Abstract

While the use of intra-day price data increases the sample size substantially for asset allocation, the usual realized covariance matrix still suffers from bias contributed from the extreme eigenvalues when the number of assets is large. We introduce a novel nonlinear shrinkage estimator for the integrated volatility matrix which shrinks the extreme eigenvalues of a realized covariance matrix back to acceptable level, and enjoys a certain asymptotic efficiency at the same time, all at a high dimensional setting where the number of assets can have the same order as the number of data points. Compared to a time-variation adjusted realized covariance estimator and the usual realized covariance matrix, our estimator demonstrates favorable performance in both simulations and a real data analysis in portfolio allocation. This include a novel maximum exposure bound and an actual risk bound when our estimator is used in constructing the minimum variance portfolio.

Key words and phrases. High dimension; Intra-day volatility; Realized covariance; Extreme eigenvalues; Portfolio allocation; Maximum exposure bound.

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1 Introduction

With the easily obtainable intra-day trading data nowadays, financial market analysts and academic researchers enjoy more accurate return or volatility matrix estimation through the substantial increase in sample size. Yet, with respect to the integrated covariance matrix estimation for asset returns, there are several well-known challenges using such intra-day price data. For instance, when tick-by-tick price data is used, the contamination by market microstructure noise (Aït-Sahalia et al., 2005, Asparouhova et al., 2013) can hugely bias the realized covariance matrix. Non-synchronous trading times presents another challenge when there are more than one asset to consider.

To present a further challenge, it is well-documented that with independent and identically distributed random vectors, random matrix theories imply that there are biased extreme eigenvalues for the corresponding sample covariance matrix when the dimension of the random vectors p has the same order as the sample size n , i.e., $p/n \rightarrow c > 0$ for some constant $c > 0$. See for instance Bai and Silverstein (2010) for more details. This suggests that the realized covariance matrix, which is essentially a sample covariance matrix when all covolatilities are constants and all log prices have zero drift with equally-spaced observation times (see the diffusion process for the log price defined in (2.1) for more details), can have biased extreme eigenvalues under the high dimensional setting $p/n \rightarrow c > 0$. The resulting detrimental effects to risk estimation or portfolio allocation are thoroughly demonstrated in Bai et al. (2009) when inter-day price data is used.

To rectify this bias problem, many researchers focused on regularized estimation of covariance or precision matrices with special structures. These go from banded (Bickel and Levina, 2008b) or sparse covariance matrix (Bickel and Levina, 2008a, Cai and Zhou, 2012, Lam and Fan, 2009, Rothman et al., 2008), sparse precision matrix (Friedman et al., 2008, Meinshausen and Bühlmann, 2006), sparse modified Cholesky factor (Pourahmadi, 2007), to a spiked covariance matrix from a factor model (Fan et al., 2008, 2011), or combinations of these (Fan et al., 2013).

Recently, Ledoit and Wolf (2012) proposed a nonlinear shrinkage formula for shrinking the extreme eigenvalues in a sample covariance matrix without assuming a particular structure of the true covariance matrix. The method is generalized in Ledoit and Wolf (2014) for portfolio allocation with remarkable results. However, such a nonlinear shrinkage formula is only applicable to the independent and identically distributed random vector setting. It is not applicable to intra-day price data since the volatility within a trading day is highly variable, so that asset returns at different time periods, albeit independent theoretically, are not identically distributed.

Lam (2016a) proves that by splitting the data into two independent portions of certain sizes, one can

achieve the same nonlinear shrinkage asymptotically without the need to evaluate a shrinkage formula as in Ledoit and Wolf (2012), which can be computationally expensive. At the same time, such a data splitting approach can be generalized to adapt to different data settings. In this paper, we modify the method proposed in Lam (2016a) to achieve nonlinear shrinkage of eigenvalues in the realized covariance matrix using intra-day price data. We use the same assumption as in Zheng and Li (2011) (see Assumption 1 in Section 2 and the details therein) to overcome the difficulty of time-varying volatilities for all underlying stocks. Ultimately, our method produces a positive definite integrated covariance matrix asymptotically almost surely with shrinkage of eigenvalues achieved nonlinearly, while local integrated covolatilities are adapted and estimated accurately. Our method is fast since it involves only eigen-decompositions of matrices of size $p \times p$, which is not computationally expensive when p is of order of hundreds. This is usually the typical order for p in the case of portfolio allocation. We also present the maximum exposure bound and the actual risk bound for portfolio allocation using our estimator as an input for the minimum variance portfolio. These bounds are important in practice as seen in the real data analysis results in Section 4.2, when our portfolio do not over-invest in individual assets, and the actual risk is small compared to other methods.

The rest of the paper is organized as follows. We first present the framework for the data together with the notations and the main assumptions to be used in Section 2. Our method of estimation is detailed in Section 2.1, while Section 3 presents all related theories. Simulation results are given in Section 4, with the theorem concerning the maximum exposure bound and the actual risk bound in portfolio allocation using our method presented in Section 4.1. A real data example of portfolio allocation is presented in Section 4.2. All proofs are presented in the Appendix.

2 Framework and Methodology

Let $\mathbf{X}_t = (X_t^{(1)}, \dots, X_t^{(p)})^\top$ be a p -dimensional log-price process which is modeled by the diffusion process

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \boldsymbol{\Theta}_t d\mathbf{W}_t, \quad t \in [0, 1], \quad (2.1)$$

where $\boldsymbol{\mu}_t$ is the drift, $\boldsymbol{\Theta}_t$ is a $p \times p$ matrix called the (instantaneous) covolatility process, and $\mathbf{W}_t = (W_t^{(1)}, \dots, W_t^{(p)})^\top$ is a p -dimensional standard Brownian motion. We want to estimate the integrated covariance matrix

$$\boldsymbol{\Sigma}_p = \int_0^1 \boldsymbol{\Theta}_t \boldsymbol{\Theta}_t^\top dt. \quad (2.2)$$

It is well-known that the log-price process $\mathbf{X}_t$ is contaminated by market microstructure noise; see Zhang (2011) for instance when the tick-by-tick high-frequency trading data is used to calculate an integrated covariance estimator. In this paper, instead of using the tick-by-tick data which has the highest observation frequency possible, we use sparsely sampled data synchronized by refresh times (Andersen et al., 2001, Barndorff-Nielsen et al., 2011), so that the theory in our paper should be readily applicable. Hence in the sequel, we assume that we can observe the price $\mathbf{X}_t$ at synchronous time points $\tau_{n,\ell}$, $\ell = 0, 1, \dots, n$. The realized covariance matrix is then defined as

$$\Sigma_p^{\text{RCV}} = \sum_{\ell=1}^n \Delta \mathbf{X}_\ell \Delta \mathbf{X}_\ell^\top, \quad \text{where } \Delta \mathbf{X}_\ell := \mathbf{X}_{\tau_{n,\ell}} - \mathbf{X}_{\tau_{n,\ell-1}}. \quad (2.3)$$

Jacod and Protter (1998) shows that as n goes to infinity, the above estimator converges weakly to the true one defined in (2.2). Hence the realized covariance matrix is one of the most frequently used estimator for the integrated covariance matrix.

While the intra-day volatility can change hugely within a short time period, it is not unreasonable to assume that the correlation of any two price processes stays constant within such a period, say within a trading day. Following Zheng and Li (2011), for $j = 1, \dots, p$, write

$$dX_t^{(j)} = \mu_t^{(j)} + \sigma_t^{(j)} dZ_t^{(j)},$$

where $\mu_t^{(j)}, \sigma_t^{(j)}$ are assumed to be càdlàg over $[0, 1]$, and the $Z_t^{(j)}$'s are one dimensional standard Brownian motions. Both the $\sigma_t^{(j)}$'s and the $Z_t^{(j)}$'s are related to Θ_t and $\mathbf{W}_t$ in (2.1). We assume further, defining $\langle X, Y \rangle_t$ to be the quadratic covariation between the processes X and Y :

Assumption 1. *The correlation matrix process of $\mathbf{Z}_t = (Z_t^{(1)}, \dots, Z_t^{(p)})^\top$, defined by*

$$R_t = (\langle Z^{(j)}, Z^{(k)} \rangle_t / t)_{1 \leq j, k \leq p},$$

is constant and non-zero on $(0, 1]$ for each j, k . Furthermore, the correlation matrix process of $\mathbf{X}_t$, defined by

$$\left(\frac{\int_0^t \sigma_s^{(j)} \sigma_s^{(k)} d\langle Z^{(j)}, Z^{(k)} \rangle_s}{\sqrt{\int_0^t (\sigma_s^{(j)})^2 ds \cdot \int_0^t (\sigma_s^{(k)})^2 ds}} \right)_{1 \leq j, k \leq p},$$

is constant on $(0, 1]$ for each j, k .

The rest of the assumptions in this paper can be found in Section 3. We present this assumption first since following Proposition 4 in Zheng and Li (2011), the log-price process $\mathbf{X}_t$ defined in (2.1) satisfying Assumption 1 is such that there exist a càdlàg process $(\gamma_t)_{t \in [0, 1]}$ and a $p \times p$ matrix Λ satisfying $\text{tr}(\Lambda \Lambda^\top) = p$

such that

$$\Theta_t = \gamma_t \Lambda. \quad (2.4)$$

The nonlinear shrinkage estimator described in the next section is based on this property.

2.1 Nonlinear shrinkage estimator

When the dimension p is large relative to the sample size n , even for a sample covariance matrix constructed from independent and identically distributed random vectors, its extreme eigenvalues will be severely biased from the true ones (see chapter 5.2 of Bai and Silverstein (2010) for example). While various assumptions have been made on the true integrated covariance matrix like sparsity (Wang and Zou, 2010) or having a factor structure (Tao et al., 2011), in this paper we follow Ledoit and Wolf (2012) and introduce nonlinear shrinkage for regularization, which does not need a particular structural assumption on the true integrated covariance matrix itself.

However, since intra-day covariance can vary hugely within a short time period, the $\Delta \mathbf{X}_\ell$'s defined in (2.3) are not identically distributed, and hence we cannot directly apply the nonlinear shrinkage formula in Ledoit and Wolf (2012) to the realized covariance matrix in (2.3). Instead, we use the data splitting idea for nonlinear shrinkage of eigenvalues in Lam (2016a), and modify their method to accommodate the intra-day volatility change base on (2.4), which is a condition derived from Assumption 1 as proved in Zheng and Li (2011).

To this end, observe that by (2.4), the integrated covariance matrix in (2.2) can be written as $\Sigma_p = \int_0^1 \gamma_t^2 dt \cdot \Lambda \Lambda^\top$. Zheng and Li (2011) proposed a so-called Time-variation adjusted realized covariance matrix, defined as

$$\check{\Sigma}_p := \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} \check{\Phi}, \quad \text{where } \check{\Phi} := \frac{p}{n} \sum_{\ell=1}^n \frac{\Delta \mathbf{X}_\ell \Delta \mathbf{X}_\ell^\top}{\|\Delta \mathbf{X}_\ell\|^2}, \quad (2.5)$$

and $\|\cdot\|$ denotes the norm of a vector. They demonstrate that $\check{\Sigma}_p$ is a good estimator for Σ_p by showing that $\text{tr}(\Sigma_p^{\text{RCV}})/p$ is a good estimator for $\int_0^1 \gamma_t^2 dt$, while $\check{\Phi}$ is good for $\Phi = \Lambda \Lambda^\top$. Here $\check{\Phi}$ plays the role of a sample covariance matrix for estimating Φ . Hence if $p/n \rightarrow c > 0$, then $\check{\Phi}$ suffers from bias to the extreme eigenvalues as well.

Remark 1. *An intuition of why $\check{\Phi}$ is similar to a sample covariance matrix can be seen as follows. If $\mu_t = \mathbf{0}$ in (2.1) and the $\tau_{n,\ell}$'s are independent of $\mathbf{W}_t$ (see Assumptions 2 and 3 respectively in Section 3), then by model (2.1), we can write*

$$\Delta \mathbf{X}_\ell = \int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t \Lambda d\mathbf{W}_t \stackrel{d}{=} \left(\int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt \right)^{1/2} \Phi^{1/2} \mathbf{Z}_\ell,$$

where $\stackrel{d}{=}$ stands for equal in distribution, and the $\mathbf{Z}_\ell$'s are independent random vectors each with $\mathbf{Z}_\ell \sim N(\mathbf{0}, \mathbf{I}_p)$. Then

$$\check{\Phi} \stackrel{d}{=} \frac{1}{n} \sum_{\ell=1}^n \frac{\Phi^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \Phi^{1/2}}{\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p} = \Phi^{1/2} \left(\frac{1}{n} \sum_{\ell=1}^n \frac{\mathbf{Z}_\ell \mathbf{Z}_\ell^\top}{\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p} \right) \Phi^{1/2}.$$

We can actually show that $\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p$ goes to 1 almost surely, leaving the above being the sample covariance matrix constructed from the $\mathbf{Z}_\ell$'s sandwiched by $\Phi^{1/2}$.

Following Lam (2016a), since the $\Delta \mathbf{X}_\ell$'s are independent following model (2.1), we split the data $\Delta \mathbf{X} = (\Delta \mathbf{X}_1, \dots, \Delta \mathbf{X}_n)$ into two independent parts, say $\Delta \mathbf{X} = (\Delta \mathbf{X}^1, \Delta \mathbf{X}^2)$, with $\Delta \mathbf{X}^i$ having size $p \times n_i$ for $i = 1, 2$, such that $n = n_1 + n_2$. Define

$$\tilde{\Phi}_i = \frac{p}{n_i} \sum_{\ell \in I_i} \frac{\Delta \mathbf{X}_\ell \Delta \mathbf{X}_\ell^\top}{\|\Delta \mathbf{X}_\ell\|^2}, \quad (2.6)$$

where $I_i = \{\ell : \Delta \mathbf{X}_\ell \in \Delta \mathbf{X}^i\}$. Carrying out an eigen-analysis on $\tilde{\Phi}_1$ defined in (2.6) above, suppose $\tilde{\Phi}_1 = \mathbf{P}_1 \mathbf{D}_1 \mathbf{P}_1^\top$. Then we introduce our estimator as

$$\hat{\Sigma}_p := \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} \hat{\Phi}, \quad \text{where } \hat{\Phi} := \mathbf{P}_1 \text{diag}(\mathbf{P}_1^\top \tilde{\Phi}_2 \mathbf{P}_1) \mathbf{P}_1^\top, \quad (2.7)$$

with $\text{diag}(\cdot)$ setting all non-diagonal elements of a matrix to 0. The estimator $\hat{\Phi}$ above belongs to a class of rotation equivariant estimator $\Phi(\mathbf{D}) = \mathbf{P}_1 \mathbf{D} \mathbf{P}_1^\top$, where $\mathbf{D}$ is a diagonal matrix, and $\mathbf{P}_1$ is the matrix containing all the eigenvectors of $\tilde{\Phi}_1$. The choice of $\mathbf{D} = \text{diag}(\mathbf{P}_1^\top \tilde{\Phi}_2 \mathbf{P}_1)$ comes from solving

$$\min_{\mathbf{D}} \|\mathbf{P}_1 \mathbf{D} \mathbf{P}_1^\top - \tilde{\Phi}_2\|_F,$$

where $\|\mathbf{A}\|_F = \text{tr}^{1/2}(\mathbf{A} \mathbf{A}^\top)$ is the Frobenius norm of a matrix. Similar to Lam (2016a), regularization of the eigenvalues in $\mathbf{D} = \text{diag}(\mathbf{P}_1^\top \tilde{\Phi}_2 \mathbf{P}_1)$ comes from the independence between $\mathbf{P}_1$ and $\tilde{\Phi}_2$, since $\Delta \mathbf{X}^1$ is independent of $\Delta \mathbf{X}^2$.

3 Asymptotic Theory and Practical Implementation

We introduce two more assumptions needed for our results to hold. Assumption 1 is presented in Section 2.

Assumption 2. *The drift in (2.1) satisfies $\boldsymbol{\mu}_t = \mathbf{0}$ for $t \in [0, 1]$. All eigenvalues of $\boldsymbol{\Theta}_t \boldsymbol{\Theta}_t^\top$ are bounded uniformly from 0 and infinity in $t \in [0, 1]$.*

Assumption 3. The observation times $\tau_{n,\ell}$'s are independent of the log-price $\mathbf{X}_t$, and there exists a constant $C > 0$ such that for all positive integer n ,

$$\max_{1 \leq \ell \leq n} n(\tau_{n,\ell} - \tau_{n,\ell-1}) \leq C.$$

We set $\boldsymbol{\mu}_t = \mathbf{0}$ in Assumption 2 for the ease of proofs and presentation. If $\boldsymbol{\mu}_t$ is slowly varying locally, the results to be presented are still valid at the expense of longer and more complex proofs. The uniform bounds on the eigenvalues of $\boldsymbol{\Theta}_t \boldsymbol{\Theta}_t^T$ are needed so that individual volatility process for each $X_t^{(i)}$ are bounded uniformly. Also, $\int_0^1 \gamma_t^2 dt > 0$ uniformly, and finally, $\|\boldsymbol{\Sigma}_p\| = O(1)$ uniformly as a result, which are all needed for our results to hold. These assumptions essentially treat γ_t as non-random. Extension to γ_t being stochastic can follow the lines of Zheng and Li (2011), but we keep it non-random for the ease of presentation and proofs as well.

Lemma 1. Let Assumptions 1, 2 and 3 hold for the log-price process $\mathbf{X}_t$ in (2.1). Then for the estimator $\hat{\boldsymbol{\Phi}}$ in (2.7), writing $\mathbf{P}_1 = (\mathbf{p}_{11}, \dots, \mathbf{p}_{1p})$, if $p/n \rightarrow c > 0$ and $\sum_{n_2 \geq 1} pn_2^{-5} < \infty$, we have

$$\max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \boldsymbol{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \boldsymbol{\Phi} \mathbf{p}_{1i}} \right| \xrightarrow{a.s.} 0,$$

where $\xrightarrow{a.s.}$ represents almost sure convergence.

Since the eigenvalues of $\hat{\boldsymbol{\Phi}}$ are the $\mathbf{p}_{1i}^T \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i}$'s, the above Lemma shows that they are regularized to $\mathbf{p}_{1i}^T \boldsymbol{\Phi} \mathbf{p}_{1i}$ asymptotically almost surely, which has values bounded by $\lambda_{\min}(\boldsymbol{\Phi})$ and $\lambda_{\max}(\boldsymbol{\Phi})$, the minimum and maximum eigenvalues of $\boldsymbol{\Phi}$ respectively. Assumption 2 ensures that these eigenvalues are uniformly bounded away from 0 and infinity, and hence $\hat{\boldsymbol{\Phi}}$ is asymptotically almost surely positive definite. This is true even when the constant $c > 1$, i.e., when p is larger than n as they grow together to infinity.

With this result, we can present the following theorem.

Theorem 1. Let all the assumptions in Lemma 1 hold. Then as $p, n \rightarrow \infty$ such that $p/n \rightarrow c > 0$, $\hat{\boldsymbol{\Sigma}}_p$ defined in (2.7) is almost surely positive definite.

This is an important result since $\boldsymbol{\Sigma}_p$ is always assumed to be positive definite, and we want our estimator to be so too. This is certainly not the case for a sample covariance matrix when $p > n$, and is still not the case for $\tilde{\boldsymbol{\Sigma}}_p$ defined in (2.5) by Zheng and Li (2011), which is demonstrated in our simulation results in Section 4.

Remark 2. Both Lemma 1 and Theorem 1 requires $\sum_{n_2 \geq 1} pn_2^{-5} < \infty$. Following Lam (2016a), we set

$n_2 = an^{1/2}$ where a is a constant, so that when $p/n \rightarrow c > 0$, the condition is satisfied. See Section 3.1 for more details on how to find n_2 with finite sample.

To present the rest of the results, we introduce a benchmark estimator for comparisons to our estimator. This estimator is called the ideal estimator, defined by

$$\mathbf{\Sigma}_{\text{ideal}} = \int_0^1 \gamma_t^2 dt \cdot \mathbf{P} \text{diag}(\mathbf{P}^T \mathbf{\Phi} \mathbf{P}) \mathbf{P}^T.$$

This is similar to the proposed estimator defined in (2.7), except that the estimator $\text{tr}(\mathbf{\Sigma}_p^{\text{RCV}})/p$ is replaced by the population counterpart $\int_0^1 \gamma_t^2 dt$, while $\tilde{\mathbf{\Phi}}_2$ is replaced by the population counterpart $\mathbf{\Phi}$. Also, $\mathbf{P}_1$ is replaced by $\mathbf{P}$, which is the matrix containing all orthonormal eigenvectors for the covariance-type matrix $\check{\mathbf{\Phi}}$ defined in (2.5) using all data points. This is in line with the ideal estimator defined in Ledoit and Wolf (2012) and Lam (2016a) which utilizes all data points for calculating the eigenmatrix $\mathbf{P}$, and assumes the knowledge of $\mathbf{\Phi}$ and $\int_0^1 \gamma_t^2 dt$. With this, we define the efficiency loss of any estimator $\hat{\mathbf{\Sigma}}$ as

$$EL(\mathbf{\Sigma}_p, \hat{\mathbf{\Sigma}}) := 1 - \frac{L(\mathbf{\Sigma}_p, \mathbf{\Sigma}_{\text{Ideal}})}{L(\mathbf{\Sigma}_p, \hat{\mathbf{\Sigma}})},$$

where $L(\mathbf{\Sigma}_p, \hat{\mathbf{\Sigma}})$ is a loss function for estimating $\mathbf{\Sigma}_p$ by $\hat{\mathbf{\Sigma}}$. We consider the Frobenius loss

$$L(\mathbf{\Sigma}_p, \hat{\mathbf{\Sigma}}) = \|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}_p\|_F^2, \quad (3.8)$$

and the inverse Stein's loss function in this paper,

$$L(\mathbf{\Sigma}_p, \hat{\mathbf{\Sigma}}) = \text{tr}(\mathbf{\Sigma}_p \hat{\mathbf{\Sigma}}^{-1}) - \log \det(\mathbf{\Sigma}_p \hat{\mathbf{\Sigma}}^{-1}) - p. \quad (3.9)$$

The class of rotation-equivariant estimator $\mathbf{\Sigma}(\mathbf{D}) = \mathbf{P} \mathbf{D} \mathbf{P}^T$ minimizes the Frobenius norm exactly at $\mathbf{\Sigma}_{\text{Ideal}}$, while similar to Proposition 2 in Lam (2016a), $\mathbf{\Sigma}_{\text{Ideal}}$ also minimizes the inverse Stein's loss within such a class of estimator. Hence it is intuitive that our estimator $\hat{\mathbf{\Sigma}}_p$ will be relatively less efficient in the sense that $EL(\mathbf{\Sigma}_p, \hat{\mathbf{\Sigma}}_p) > 0$. It turns out that asymptotically, $\hat{\mathbf{\Sigma}}_p$ is doing as good as $\mathbf{\Sigma}_{\text{Ideal}}$, as shown in the Theorem 2 below. To present this theorem, we need to make two more assumptions:

Assumption 4. Let $v_{n,1} \geq \dots \geq v_{n,p}$ be the p eigenvalues of $\mathbf{\Phi}$. Define $H_n(v) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{v_{n,i} \leq v\}}$ the empirical distribution function of the population eigenvalues. We assume $H_n(v)$ converges to some non-random limit H at every point of continuity of H .

Assumption 5. The support of H defined above is the union of a finite number of compact intervals bounded away from zero and infinity. Also, there exists a compact interval in $(0, +\infty)$ that contains the

support of H_n for each n .

These two assumptions are essentially Assumptions (A3) and (A4) in Lam (2016a) applied on Φ .

Theorem 2. *Let all the assumptions in Lemma 1 hold, together with Assumption 4 and 5. Then as $p, n \rightarrow \infty$ such that $p/n \rightarrow c > 0$, we have $EL(\Sigma_p, \hat{\Sigma}_p) \leq 0$ almost surely with respect to both the Frobenius and the inverse Stein's loss functions, as long as $p^{-1}L(\Sigma_p, \Sigma_{\text{Ideal}})$ does not tend to 0 almost surely.*

The requirement $p^{-1}L(\Sigma_p, \Sigma_{\text{Ideal}})$ not going to 0 almost surely eliminates the case $\Sigma_p = \int_0^1 \gamma_t^2 dt \cdot \mathbf{I}_p$, when both the loss functions will attain 0 for the the ideal estimator. Our estimator will still do a good job in such a case since $\text{tr}(\Sigma_p^{\text{RCV}})/p$ will still be a good estimator for $\int_0^1 \gamma_t^2 dt$ by the proof of Theorem 1, while $\hat{\Phi}$ can still do a fine job when permutation of the data is allowed as demonstrated in the simulation results in Lam (2016a). Improvement by averaging and permutation will be described in Section 3.1.

3.1 Practical Implementation

Following Assumption 1, $\Delta \mathbf{X}_\ell \Delta \mathbf{X}_\ell^\top / \|\Delta \mathbf{X}_\ell\|^2$ is independent of γ_t and is similar to a data point in constructing a sample covariance matrix, which is independent of each others for different ℓ ; see Remark 1 in Section 2.1. This observation permits us to permute the data beforehand, say at the j th permutation, we form a data matrix $\Delta \mathbf{X}^{(j)} = (\Delta \mathbf{X}_1^{(j)}, \Delta \mathbf{X}_2^{(j)})$, with $\Delta \mathbf{X}_i^{(j)}$ having size $p \times n_i$ for $i = 1, 2$, such that $n = n_1 + n_2$. Then we construct

$$\tilde{\Phi}_i^{(j)} = \frac{p}{n_i} \sum_{\ell \in I_i^{(j)}} \frac{\Delta \mathbf{X}_\ell \Delta \mathbf{X}_\ell^\top}{\|\Delta \mathbf{X}_\ell\|^2}, \quad (3.10)$$

where $I_i^{(j)} = \{\ell : \Delta \mathbf{X}_\ell \in \Delta \mathbf{X}_i^{(j)}\}$, and perform eigen-analysis on $\tilde{\Phi}_1^{(j)}$, say $\tilde{\Phi}_1^{(j)} = \mathbf{P}_1^{(j)} \mathbf{D}_1^{(j)} \mathbf{P}_1^{(j)\top}$. The we can form the j th estimator as

$$\hat{\Sigma}_p^{(j)} := \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} \hat{\Phi}^{(j)}, \quad \text{where } \hat{\Phi}^{(j)} := \mathbf{P}_1^{(j)} \text{diag}(\mathbf{P}_1^{(j)\top} \tilde{\Phi}_2^{(j)} \mathbf{P}_1^{(j)}) \mathbf{P}_1^{(j)\top}. \quad (3.11)$$

If we perform M permutations and get M estimators as above, we can define the averaged estimator as

$$\hat{\Sigma}_{p,M} := \frac{1}{M} \sum_{j=1}^M \hat{\Sigma}_p^{(j)}. \quad (3.12)$$

Note that in all M estimators, we are only using one split location, n_1 , for the data, instead of using several of them and then average the results similar to the grand average estimator in Abadir et al. (2010). To

find the best split location empirically, we minimize the following function:

$$g(m) = \left\| \frac{1}{M} \sum_{j=1}^M (\hat{\Phi}_p^{(j)} - \tilde{\Phi}_2^{(j)}) \right\|_F^2, \quad (3.13)$$

where $\tilde{\Phi}_2^{(j)}$ is defined in (3.10) and $\hat{\Phi}^{(j)}$ in (3.11). A very similar function is also used to determine the split location for nonlinear shrinkage of a covariance matrix in Lam (2016a).

We now show that the averaged estimator defined in (3.12) also enjoys good asymptotic efficiency.

Theorem 3. *Let all the assumptions in Lemma 1 hold, together with Assumption 4 and 5. Suppose the number of permutations M is finite in (3.12). Then as $p, n \rightarrow \infty$ such that $p/n \rightarrow c > 0$, we have $EL(\Sigma_p, \hat{\Sigma}_{p,M}) \leq 0$ almost surely with respect to both the Frobenius and the inverse Stein's loss functions, as long as $p^{-1}L(\Sigma_p, \hat{\Sigma}_{Ideal})$ does not tend to 0 almost surely.*

In practice, the estimator $\hat{\Sigma}_{p,M}$, with a good choice of m , performs much better than using just $M = 1$. We use $M = 50$ which provides a good trade-off between computational complexity and estimation accuracy with respect to the Frobenius or the inverse Stein's loss functions. For minimizing $g(m)$ defined in (3.13), we search the following split locations:

$$m = [2n^{1/2}, 0.2n, 0.4n, 0.6n, 0.8n, n - 2.5n^{1/2}, n - 1.5n^{1/2}].$$

Except for the case $\Sigma_p = \int_0^1 \gamma_t^2 dt \mathbf{I}_p$ which needs n to be as small as possible (see the arguments provided in Lam (2016a)), the two split locations $[n - 2.5n^{1/2}]$ and $[n - 1.5n^{1/2}]$ are those satisfying the condition $\sum_{n_2 \geq 1} pn_2^{-5} < \infty$ needed in all theorems presented when $p/n \rightarrow c > 0$. We include $0.2n$ to $0.8n$ for accommodating finite sample performance.

4 Empirical Results

We carry out simulation studies to compare the performances of our estimator in (2.7), the time variation-adjusted realized covariance matrix in (2.5) and the realized covariance matrix in (2.3) by comparing their Frobenius and inverse Stein's losses defined in (3.8) and (3.9) respectively. Then in Section 4.1, we consider a trading exercise using simulated market data and compare the risks associated with the minimum variance portfolios constructed using these three different estimators. Finally, in Section 4.2, we consider real data from the New York Stock Exchange.

Consider two different scenarios for the diffusion process $\{\mathbf{X}_t\}$ defined in (2.1), with $\boldsymbol{\mu}_t = \mathbf{0}$ and

$\Theta_t = \gamma_t \mathbf{\Lambda}$ as in (2.4). One has γ_t being piecewise constant, the other has γ_t being continuous, detailed as follows:

Design I: Piecewise constants. We take γ_t to be

$$\gamma_t = \begin{cases} \sqrt{0.0007}, & t \in [0, 1/4) \cup [3/4, 1], \\ \sqrt{0.0001}, & t \in [1/4, 3/4]. \end{cases}$$

Design II: Continuous path. We take γ_t to be

$$\gamma_t = \sqrt{0.0009 + 0.0008 \cos(2\pi t)}, t \in [0, 1].$$

We assume $\mathbf{\Lambda} = \mathbf{I}_p$ and the observation times are taken to be equidistant, where $\tau_{n,\ell} = \ell/n, \ell = 1, \dots, n$. We generate $\{\mathbf{X}_t\}$ using model (2.1) and get $n = 200$ discrete observations, and consider $p = 100, 200$. For each design and each (n, p) combination, we repeat 1000 times the simulations, and compare the mean Frobenius and inverse Stein's losses for our proposed estimator, the time variation-adjusted realized covariance matrix and the realized covariance matrix.

Table 1 presents the simulation results. It is clear that overall, our proposed estimator performs the best. In particular, since the realized covariance or the time variation-adjusted realized covariance matrices are singular when $p = 200$, their inverses do not exist. In contrast, our proposed estimator is always non-singular and stable even in this case, which is in line with Theorem 1.

Design I	Proposed	Time variation-adjusted	Realized covariance
	Frobenius loss		
$p = 100$	.13 _(.02)	2.8 _(.04)	3.6 _(.06)
$p = 200$	.55 _(.17)	69 ₍₃₁₎	1564 ₍₆₃₎
	Inverse Stein's loss		
$p = 100$	.17 _(.014)	5.63 _(.058)	7.08 _(.08)
$p = 200$	88 ₍₁₅₎	-	-
Design II	Proposed	Time variation-adjusted	Realized covariance
	Frobenius loss		
$p = 100$	.29 _(.03)	6.32 _(.09)	7.55 _(.1)
$p = 200$	.38 _(.03)	13 ₍₁₀₎	15 ₍₂₀₎
	Inverse Stein's loss		
$p = 100$	.54 _(.1)	693 ₍₃₁₎	1232 _(53.8)
$p = 200$	88 ₍₁₆₎	-	-

Table 1: Mean and standard deviation (in bracket) of losses for different methods. All values reported in this table are multiplied by 1000. Upper table: results for Design I. Lower table: results for Design II. For $p = 200$, the time variation-adjusted and realized covariance matrices are always singular, and hence inverse Stein's loss are at infinity.

4.1 A market trading exercise

As an application in finance, we simulate market trading data in this section and construct minimum variance portfolio using the three different estimators compared in the previous section. Given an integrated covariance matrix Σ_p , the minimum variance portfolio solves $\min_{\mathbf{w}: \mathbf{w}^\top \mathbf{1}_p = 1} \mathbf{w}^\top \Sigma_p \mathbf{w}$, where $\mathbf{1}_p$ is a vector of p ones. The solution to the above is given by

$$\mathbf{w}_{\text{opt}} = \frac{\Sigma_p^{-1} \mathbf{1}_p}{\mathbf{1}_p^\top \Sigma_p^{-1} \mathbf{1}_p}. \quad (4.14)$$

Before presenting our simulation settings, we present a theorem concerning the minimum variance portfolio (4.14) constructed using our integrated covariance matrix estimator $\hat{\Sigma}_{p,M}$. In the sequel, we denote $\|\cdot\|_{\max}$ the maximum absolute value of a vector, and define the condition number of a positive semi-definite matrix $\mathbf{A}$ to be $\text{Cond}(\mathbf{A}) = \lambda_{\max}(\mathbf{A})/\lambda_{\min}(\mathbf{A})$.

Theorem 4. *Let all the assumptions in Lemma 1 hold. Suppose the number of permutations M is finite in (3.12). Then as $p, n \rightarrow \infty$ such that $p/n \rightarrow c > 0$, almost surely,*

$$p^{1/2} \|\hat{\mathbf{w}}_{\text{opt}}\|_{\max} \leq \text{Cond}(\Phi), \quad pR(\hat{\mathbf{w}}_{\text{opt}}) \leq \text{Cond}^2(\Phi) \lambda_{\max}(\Sigma_p),$$

where $\hat{\mathbf{w}}_{\text{opt}}$ is the weight in (4.14) with Σ_p substituted by $\hat{\Sigma}_{p,M}$ defined in (3.12). The function $R(\hat{\mathbf{w}}_{\text{opt}}) = \hat{\mathbf{w}}_{\text{opt}}^\top \Sigma_p \hat{\mathbf{w}}_{\text{opt}}$ represents the actual risk when investing using $\hat{\mathbf{w}}_{\text{opt}}$ as the portfolio weights.

This theorem shows that the maximum absolute weight, we name it the maximum exposure of the portfolio, is decaying at a rate $p^{-1/2}$, while the actual risk is decaying at a rate p^{-1} . It is an important quality of our estimator when applied to portfolio allocation, since we do not want to over-invest into a single asset when there are many other assets for risk diversification. Such over-investment can result in huge loss if the involved asset suddenly drops in price due to random events over the investment period. At the same time, the actual risk of the portfolio is decaying linearly as we have more and more assets for diversification. We demonstrate the results in Theorem 4 in Table 2 and Section 4.2.

For simulating the price data, following Barndorff-Nielsen et al. (2011) and Fan et al. (2012), we simulate $p = 100$ stock prices for 200 days using $X_t^{(i)} = X_t^{(i)} + \epsilon_t^{(i)}$, where $X_t^{(i)}$ is the underlying log-price, and $\epsilon_t^{(i)}$ models the market microstructure noise, with $\epsilon_t^{(i)} \sim N(0, 0.0005^2)$ and are assumed to be independent of each other. The underlying log-price $X_t^{(i)}$ is generated by the stochastic volatility model. For $i = 1, \dots, 100$,

$$dX_t^{(i)} = \mu^{(i)} dt + \rho^{(i)} \sigma_t^{(i)} dB_t^{(i)} + \sqrt{1 - (\rho^{(i)})^2} \sigma_t^{(i)} dW_t + \nu^{(i)} dZ_t,$$

where $\{W_t\}$, $\{Z_t\}$ and the $\{B_t^{(i)}\}$'s are all independent standard Brownian motions. The process $\{Z_t\}$ plays the role of a pervasive factor, which is usually the market factor in asset returns. The spot volatility $\sigma_t^{(i)} = \exp(\varrho_t^{(i)})$ follows the independent Ornstein-Uhlenbeck process

$$d\varrho_t^{(i)} = \alpha^{(i)}(\beta_0^{(i)} - \varrho_t^{(i)})dt + \beta_1^{(i)}dU_t^{(i)},$$

where the $\{U_t^{(i)}\}$'s are independent standard Brownian motions. We use $(\mu^{(i)}, \beta_0^{(i)}, \beta_1^{(i)}, \alpha^{(i)}, \rho^{(i)}) = (0.03x_1^{(i)}, -x_2^{(i)}, 0.75x_3^{(i)}, -1/40x_4^{(i)}, -0.7)$ and $\nu^{(i)} = \exp(\beta_0^{(i)})$, where the $x_j^{(i)}$'s are independent and uniformly distributed on the interval $[0.7, 1.3]$. The initial value of each log-price is set at $X_0^{(i)} = 1$ and the starting spot volatility $\varrho_0^{(i)} = 0$.

We simulate the trading times independently from the price data assuming the transaction times for each stock follow independent Poisson processes with rates $\lambda_1, \dots, \lambda_{100}$ respectively, where $\lambda_i = 0.01i \times 23400$. We set this because normal trading time for one day is 23400 seconds.

After simulating the data, we split a trading day into 15-minute intervals, and set the price data for each stock at the end of each interval as the price observed at the trade right before the end of the interval. The data is used to calculate various integrated covariance estimators, including our proposed one.

At the start, we invest 1 unit of capital using the minimum variance allocations (4.14) constructed from using different estimators of the integrated covariance matrix. Each time we use a 60-day training window (so the first trade starts on day 61, and the last one on day 200) and we re-evaluate our portfolio weights every 5 days, using the past 60 days of data as a training set, until we reach day 195.

In Table 2, we report the mean of three risks. The first one is the theoretical risk $R(\mathbf{w}_{\text{opt}}) = \mathbf{w}_{\text{opt}}^T \mathbf{\Sigma}_p \mathbf{w}_{\text{opt}}$, where $\mathbf{w}_{\text{opt}}$ is calculated as in (4.14) using the true integrated covariance matrix of the underlying log-return over the past 60-day training period and $\mathbf{\Sigma}_p$ is the true integrated covariance matrix over the 5-day investment period. The second one is the actual risk $R(\hat{\mathbf{w}}_{\text{opt}}) = \hat{\mathbf{w}}_{\text{opt}}^T \mathbf{\Sigma}_p \hat{\mathbf{w}}_{\text{opt}}$, where $\hat{\mathbf{w}}_{\text{opt}}$ is calculated using different integrated covariance matrix estimators. Finally the perceived risk is defined by $\hat{R}(\hat{\mathbf{w}}_{\text{opt}}) = \hat{\mathbf{w}}_{\text{opt}}^T \hat{\mathbf{\Sigma}}_p \hat{\mathbf{w}}_{\text{opt}}$.

We can see from Table 2 that our method has the best performance among all three different methods, and has the risk closet to the theoretical one. In particular, our method has the smallest actual risk, which is the most relevant risk in practice. Such a small actual risk for our method is also consistent with the actual risk bound in Theorem 4, when the number of assets is $p = 100$ which is relatively large.

Theoretical risk	.735		
	Proposed	Time variation-adjusted	Realized covariance
Actual risk	.922	3.918	4.115
Perceived risk	.753	3.869	4.034

Table 2: Mean of theoretical risk $R(\mathbf{w}_{\text{opt}})$, actual risk $R(\hat{\mathbf{w}}_{\text{opt}})$ and perceived risk $\hat{R}(\hat{\mathbf{w}}_{\text{opt}})$.

4.2 Portfolio allocation on NYSE data

We consider $p = 45$ stocks from the New York Stock Exchange from January 1 of 2013 to December 31 of 2013 (245 trading days). We choose the stocks from mid-cap energy sector stocks. We downloaded all the trades of these stocks from Wharton Research Data Services (WRDS, <https://wrds-web.wharton.upenn.edu/>). The raw data are of high frequency nature. As mentioned before, the stocks have non-synchronous trading times and all the log-prices are contaminated by market microstructure noise.

Like the market trading exercise in Section 4.1, we consider trades in 15-minute intervals on every trading day from 9:30 to 16:00, with each log-price being the observed one from a trade right before a 15-minute interval ends. This results in a total of 6732 observations over the 245 trading days. Hence on average there are around 27 observations per day.

We consider two settings. For the first one, we consider 20-day training windows and re-evaluate portfolio weights every 5 days. Another setting use 5-day training windows and re-evaluate portfolio weights everyday. We use the annualized out-of-sample standard deviation $\hat{\sigma}$, together with the annualized portfolio return $\hat{\mu}$ and the Sharpe ratio $\hat{\mu}/\hat{\sigma}$ to gauge the performance of each method. For 20-day training windows and 5 day re-evaluation period, $\hat{\mu}$ and $\hat{\sigma}$ are defined by

$$\hat{\mu} = 52 \times \frac{1}{45} \sum_{i=5}^{49} \mathbf{w}_i^T \mathbf{r}_i, \quad \hat{\sigma} = \left(52 \times \frac{1}{45} \sum_{i=5}^{49} (\mathbf{w}_i^T \mathbf{r}_i - \hat{\mu})^2 \right)^{1/2}.$$

We use the annualized out-of-sample standard deviation since we do not know the true underlying integrated covariance matrix, and hence the actual risk cannot be calculated. For 5-day training windows with daily re-evaluation of portfolio weights, $\hat{\mu}$ and $\hat{\sigma}$ are defined by

$$\hat{\mu} = 252 \times \frac{1}{240} \sum_{i=6}^{245} \mathbf{w}_i^T \mathbf{r}_i, \quad \hat{\sigma} = \left(252 \times \frac{1}{240} \sum_{i=6}^{245} (\mathbf{w}_i^T \mathbf{r}_i - \hat{\mu})^2 \right)^{1/2}.$$

On top of all the above, we also report the mean maximum exposure $\|\hat{\mathbf{w}}_{\text{opt}}\|_{\max}$ over all investment periods for the portfolios constructed under different methods.

Table 3 shows the results. For both settings, we see that the annualized out-of-sample standard deviation is the smallest for our method. The difference between different methods for the setting with

20-day training window Re-evaluate every 5 days	Proposed (%)	Time variation adjusted(%)	Realized covariance(%)
Annualized return	5.50	5.87	10.59
Annualized out-of-sample SD	11.07	11.16	12.67
Sharpe ratio	49.66	52.62	83.59
Mean maximum exposure	15.19 _(3.26)	19.82 _(4.74)	28.86 _(11.05)

5-day training window Re-evaluate everyday	Proposed (%)	Time variation adjusted(%)	Realized covariance(%)
Annualized return	7.40	8.25	7.39
Annualized out-of-sample SD	10.78	11.66	13.89
Sharpe ratio	68.69	73.92	53.19
Mean maximum exposure	11.47 _(2.19)	24.06 _(7.11)	31.27 _(18.76)

Table 3: Percentage annualized return, out-of-sample standard deviation, Sharpe ratio and mean maximum exposure (standard deviation in bracket) for different methods. Upper table: Use 20-day training windows and re-evaluate portfolio weights every 5 days. Lower table: Use 5-day training windows and re-evaluate portfolio weights everyday.

20-day training windows is less pronounced. This is because for a 20-day window, we have on average $27 \times 20 = 540$ data points for calculating different integrated covariance estimators. The time variation-adjusted and realized covariance matrices will not be too much disadvantaged since the bias problem from having high dimension is not too serious as we have $p/n = 45/540 = 0.083$. On the other hand, the second setting has on average $27 \times 5 = 135$ data points and so $p/n = 45/135 = 0.33$, so that the bias from having high dimension for the two estimators is much more serious than the first case. Our estimator has clearly done a good job in minimizing the risk in both situations. It would seem that we can solve the problem simply by using a longer training window. However, the problem of using a longer training window is that the true integrated covariance matrix from the training period will likely to be different from the investment period ahead if such a window is too long.

Observe also the mean maximum exposure for our method is always smaller than other methods, with smaller standard deviations. For the case of 5-day training window, the differences in maximum exposure among the three different methods are huge, with our method having this at 11.5% which is reasonable, and compatible to the results in Theorem 4. At around 25% or even 30% for the other two methods, there is a much bigger risk of suffering a loss due to random events from the heavily invested individual assets.

5 Appendix

We present the proofs of all lemmas and theorems in this section.

Proof of Lemma 1. Using Remark 1, we can write

$$\Delta \mathbf{X}_\ell = \left(\int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt \right)^{1/2} \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell,$$

where the $\mathbf{Z}_\ell$'s are independent of each other and each $\mathbf{Z}_\ell$ is a p dimensional vector with independent standard normal entries. Then we can decompose

$$\frac{\mathbf{p}_{1i}^\top \tilde{\mathbf{\Phi}} \mathbf{p}_{1i} - \mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}} = I_{i1} + I_{i2},$$

where

$$I_{i1} = \frac{\mathbf{p}_{1i}^\top \tilde{\mathbf{\Phi}} \mathbf{p}_{1i} - n_2^{-1} \sum_{\ell \in I_2} (\mathbf{p}_{1i}^\top \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell)^2}{\mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}}, \quad I_{i2} = \frac{n_2^{-1} \sum_{\ell \in I_2} (\mathbf{p}_{1i}^\top \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell)^2 - \mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}}.$$

First of all, since we can write

$$n_2^{-1} \sum_{\ell \in I_2} (\mathbf{p}_{1i}^\top \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell)^2 = \mathbf{p}_{1i}^\top \left(n_2^{-1} \sum_{\ell \in I_2} \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \mathbf{\Phi}^{1/2} \right) \mathbf{p}_{1i},$$

with $n_2^{-1} \sum_{\ell \in I_2} \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \mathbf{\Phi}^{1/2}$ a proper sample covariance matrix for estimating $\mathbf{\Phi}$, we can use Lemma 1 of Lam (2016a) to conclude that

$$\max_{1 \leq i \leq p} |I_{i2}| \xrightarrow{a.s.} 0.$$

Hence it remains to show that $\max_{1 \leq i \leq p} |I_{i1}| \xrightarrow{a.s.} 0$. To this end, consider

$$\begin{aligned} \max_{1 \leq i \leq p} |I_{i1}| &= \max_{1 \leq i \leq p} \left| \frac{n_2^{-1} \sum_{\ell \in I_2} \frac{(\mathbf{p}_{1i}^\top \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell)^2}{\mathbf{Z}_\ell^\top \mathbf{\Phi} \mathbf{Z}_\ell / p} - \mathbf{p}_{1i}^\top \left(n_2^{-1} \sum_{\ell \in I_2} \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \mathbf{\Phi}^{1/2} \right) \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}} \right| \\ &= \max_{1 \leq i \leq p} \left| \frac{n_2^{-1} \sum_{\ell \in I_2} \left(\frac{1}{\mathbf{Z}_\ell^\top \mathbf{\Phi} \mathbf{Z}_\ell / p} - 1 \right) \mathbf{p}_{1i}^\top \mathbf{\Phi}^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \mathbf{\Phi}^{1/2} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \mathbf{\Phi} \mathbf{p}_{1i}} \right| \\ &\leq \max_{\ell \in I_2} \left| \frac{1}{\mathbf{Z}_\ell^\top \mathbf{\Phi} \mathbf{Z}_\ell / p} - 1 \right| \cdot \left(1 + \max_{1 \leq i \leq p} |I_{i2}| \right) \xrightarrow{a.s.} 0, \end{aligned}$$

if we can show further that $\max_{\ell \in I_2} \left| \frac{1}{\mathbf{Z}_\ell^\top \mathbf{\Phi} \mathbf{Z}_\ell / p} - 1 \right| \xrightarrow{a.s.} 0$.

To show this, using Lemma 2.7 of Bai and Silverstein (1998), we have

$$E(\mathbf{Z}_\ell^\top \mathbf{\Phi} \mathbf{Z}_\ell - \text{tr}(\mathbf{\Phi}))^6 \leq K_6 (E^3 |z_{\ell,1}|^4 \text{tr}^3(\mathbf{\Phi}^2) + E |z_{\ell,1}|^{12} \text{tr}(\mathbf{\Phi}^6)),$$

where K_6 is a constant independent of ℓ , n and p . This implies that, since $\text{tr}(\Phi) = p$,

$$\begin{aligned} E\left(\max_{\ell \in I_2} \left| \frac{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - 1 \right|^6\right) &\leq n_2 \cdot K_6 \left(E^3 |z_{\ell,1}|^4 \frac{\text{tr}^3(\Phi^2)}{p^6} + E |z_{\ell,1}|^{12} \frac{\text{tr}(\Phi^6)}{p^6} \right) \\ &= O(n_2 p^{-3}). \end{aligned}$$

The rate in the last line comes from Assumption 2 that $\Theta_t \Theta_t^T = \gamma_t^2 \Phi$ has all its eigenvalues uniformly bounded away from 0 and infinity, so that $\text{tr}^3(\Phi^2) = O(p^3)$ and $\text{tr}(\Phi^6) = O(p)$, and the fact that the higher order moments of the $z_{\ell,1}$'s are all finite since they are all normally distributed.

Finally, since $n_2 = O(n^{1/2})$ and p has the same order as n , we have $O(n_2 p^{-3}) = O(n^{-5/2})$. Since $\sum_{n \geq 1} n^{-5/2} < \infty$, through the Borel-Cantelli lemma, we have proved that $\max_{\ell \in I_2} \left| \frac{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - 1 \right| \xrightarrow{a.s.} 0$, meaning that $\max_{\ell \in I_2} \left| \frac{1}{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell / p} - 1 \right| \xrightarrow{a.s.} 0$ as well. This completes the proof of the lemma. $\square$

Proof of Theorem 1. Firstly, by Lemma 1, $\hat{\Phi}$ defined in (2.7) is almost surely positive definite since all its eigenvalues are almost surely $\mathbf{p}_{1i}^T \Phi \mathbf{p}_{1i}$ for some i as $n, p \rightarrow \infty$ such that $p/n \rightarrow c > 0$, and Assumption 1 ensures that these values are uniformly bounded away from 0 and infinity. Hence, the proof of the theorem is complete if we can show that $\text{tr}(\Sigma_p^{\text{RCV}})/p > 0$ uniformly almost surely.

To this end, consider

$$\begin{aligned} \left| \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} - \int_0^1 \gamma_t^2 dt \right| &= \left| \frac{\sum_{\ell=1}^n \Delta \mathbf{X}_\ell^T \Delta \mathbf{X}_\ell}{p} - \int_0^1 \gamma_t^2 dt \right| \\ &= \left| \frac{\sum_{\ell=1}^n \int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt \cdot \mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - \int_0^1 \gamma_t^2 dt \right| \\ &\leq \max_{1 \leq \ell \leq n} \left| \frac{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - 1 \right| \int_0^1 \gamma_t^2 dt. \end{aligned}$$

From the proof of $\max_{\ell \in I_2} \left| \frac{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - 1 \right| \xrightarrow{a.s.} 0$ in the last part of the proof of Lemma 1, we can replace n_2 there by n and conclude that

$$\begin{aligned} E\left(\max_{1 \leq \ell \leq n} \left| \frac{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - 1 \right|^6\right) &\leq n \cdot K_6 \left(E^3 |z_{\ell,1}|^4 \frac{\text{tr}^3(\Phi^2)}{p^6} + E |z_{\ell,1}|^{12} \frac{\text{tr}(\Phi^6)}{p^6} \right) \\ &= O(np^{-3}) = O(n^{-2}). \end{aligned}$$

Since $\sum_{n \geq 1} n^{-2} < \infty$, we have proved that $\max_{1 \leq \ell \leq n} \left| \frac{\mathbf{Z}_\ell^T \Phi \mathbf{Z}_\ell}{p} - 1 \right| \xrightarrow{a.s.} 0$. This shows that $\text{tr}(\Sigma_p^{\text{RCV}})/p \xrightarrow{a.s.} \int_0^1 \gamma_t^2 dt$, which is uniformly larger than 0 by Assumption 1. This completes the proof of the theorem. $\square$

To prove Theorem 2, we first present the following lemma and its proof.

Lemma 2. *Let all the assumptions in Lemma 1 hold, together with Assumption 4 and 5. Denote by $v_1^{(1)} \geq \dots \geq v_p^{(1)}$ the eigenvalues of $\tilde{\Phi}_1$ defined in (2.6) with corresponding eigenvectors $\mathbf{p}_{11}, \dots, \mathbf{p}_{1p}$, and $v_1 \geq \dots \geq v_p$ the eigenvalues of $\check{\Phi}$ defined in (2.5) with corresponding eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_p$. Then there exist positive functions $\delta_1(\cdot) = \delta(\cdot)$ and distribution functions $F_1 = F$ such that*

$$\begin{aligned} p^{-1} \sum_{j=1}^p \mathbf{1}_{\{x \geq v_j^{(1)}\}} &\xrightarrow{a.s.} F_1(x), \quad p^{-1} \sum_{j=1}^p \mathbf{1}_{\{x \geq v_j\}} \xrightarrow{a.s.} F(x), \\ p^{-1} \sum_{j=1}^p \mathbf{p}_{1i}^\top \Phi \mathbf{p}_{1i} \mathbf{1}_{\{x \geq v_j^{(1)}\}} &\xrightarrow{a.s.} \int_{-\infty}^x \delta_1(\lambda) dF_1(\lambda), \quad p^{-1} \sum_{j=1}^p \mathbf{p}_i^\top \Phi \mathbf{p}_i \mathbf{1}_{\{x \geq v_j\}} \xrightarrow{a.s.} \int_{-\infty}^x \delta(\lambda) dF(\lambda). \end{aligned}$$

We do not write down the explicit form of the functions $\delta_1(\cdot)$ and $\delta(\cdot)$ since they are not important for the proof of any subsequent theorems. Interested readers are referred to equation (2.7) and (2.9) of Lam (2016a).

Proof of Lemma 2. Write $\Delta \mathbf{X}_\ell = \left(\int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt \right)^{1/2} \Phi^{1/2} \mathbf{Z}_\ell$ as in Remark 1. Define for $i = 1, 2$,

$$\tilde{\Phi}_{i,\text{sam}} = n_i^{-1} \sum_{\ell \in I_i} \Phi^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \Phi^{1/2}, \quad \check{\Phi}_{\text{sam}} = n^{-1} \sum_{\ell=1}^n \Phi^{1/2} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \Phi^{1/2},$$

which are all proper sample covariance matrices. Let $v_{1,\text{sam}}^{(i)} \geq \dots \geq v_{p,\text{sam}}^{(i)}$ be the eigenvalues of $\tilde{\Phi}_{i,\text{sam}}$ with corresponding eigenvectors $\mathbf{p}_{i1,\text{sam}}, \dots, \mathbf{p}_{ip,\text{sam}}$. Also, let $v_{1,\text{sam}} \geq \dots \geq v_{p,\text{sam}}$ be the eigenvalues of $\check{\Phi}_{\text{sam}}$ with corresponding eigenvectors $\mathbf{p}_{1,\text{sam}}, \dots, \mathbf{p}_{p,\text{sam}}$. Suppose we are able to show the following. For any $z \in \mathbb{C}^+$,

$$\begin{aligned} p^{-1} \text{tr}((\tilde{\Phi}_{1,\text{sam}} - z \mathbf{I}_p)^{-1}) - p^{-1} \text{tr}((\tilde{\Phi}_1 - z \mathbf{I}_p)^{-1}) &\xrightarrow{a.s.} 0, \\ p^{-1} \text{tr}((\check{\Phi}_{\text{sam}} - z \mathbf{I}_p)^{-1}) - p^{-1} \text{tr}((\check{\Phi} - z \mathbf{I}_p)^{-1}) &\xrightarrow{a.s.} 0, \\ p^{-1} \text{tr}((\tilde{\Phi}_{1,\text{sam}} - z \mathbf{I}_p)^{-1} \Phi) - p^{-1} \text{tr}((\tilde{\Phi}_1 - z \mathbf{I}_p)^{-1} \Phi) &\xrightarrow{a.s.} 0, \\ p^{-1} \text{tr}((\check{\Phi}_{\text{sam}} - z \mathbf{I}_p)^{-1} \Phi) - p^{-1} \text{tr}((\check{\Phi} - z \mathbf{I}_p)^{-1} \Phi) &\xrightarrow{a.s.} 0. \end{aligned} \tag{5.15}$$

The above can in fact be written as differences of Stieltjes transforms of certain nondecreasing functions. The differences in their inverse Stieltjes transforms must then converge to 0 almost surely as well, i.e., at

the point of continuity x of these nondecreasing functions,

$$\begin{aligned}
& p^{-1} \sum_{j=1}^p \mathbf{1}_{\{x \geq v_{j,\text{sam}}^{(1)}\}} - p^{-1} \sum_{j=1}^p \mathbf{1}_{\{x \geq v_j^{(1)}\}} \xrightarrow{a.s.} 0, \\
& p^{-1} \sum_{j=1}^p \mathbf{1}_{\{x \geq v_{j,\text{sam}}\}} - p^{-1} \sum_{j=1}^p \mathbf{1}_{\{x \geq v_j\}} \xrightarrow{a.s.} 0, \\
& p^{-1} \sum_{j=1}^p \mathbf{p}_{1j,\text{sam}}^\top \Phi \mathbf{p}_{1j,\text{sam}} \mathbf{1}_{\{x \geq v_{j,\text{sam}}^{(1)}\}} - p^{-1} \sum_{j=1}^p \mathbf{p}_{1j}^\top \Phi \mathbf{p}_{1j} \mathbf{1}_{\{x \geq v_j^{(1)}\}} \xrightarrow{a.s.} 0, \\
& p^{-1} \sum_{j=1}^p \mathbf{p}_{j,\text{sam}}^\top \Phi \mathbf{p}_{j,\text{sam}} \mathbf{1}_{\{x \geq v_{j,\text{sam}}\}} - p^{-1} \sum_{j=1}^p \mathbf{p}_j^\top \Phi \mathbf{p}_j \mathbf{1}_{\{x \geq v_j\}} \xrightarrow{a.s.} 0.
\end{aligned} \tag{5.16}$$

Interested readers are referred to Lam (2016a) or Ledoit and P  ch   (2011) for the definitions of Stieltjes transform and its inversion. Theorem 4 of Ledoit and P  ch   (2011) indicates that for all $x \in \mathbb{R}$, there exist functions $\delta_1(\cdot)$ and $\delta(\cdot)$ with corresponding distribution functions F_1 and F such that

$$\begin{aligned} p^{-1} \sum_{j=1}^p \mathbf{1}_{\{\lambda \geq v_{j,\text{sam}}^{(1)}\}} &\xrightarrow{a.s.} F_1(\lambda), \\ p^{-1} \sum_{j=1}^p \mathbf{1}_{\{\lambda \geq v_{j,\text{sam}}\}} &\xrightarrow{a.s.} F(\lambda), \\ p^{-1} \sum_{j=1}^p \mathbf{p}_{1j,\text{sam}}^\top \Phi \mathbf{p}_{1j,\text{sam}} \mathbf{1}_{\{x \geq v_{j,\text{sam}}^{(1)}\}} &\xrightarrow{a.s.} \int_{-\infty}^x \delta_1(\lambda) dF(\lambda), \\ p^{-1} \sum_{j=1}^p \mathbf{p}_{j,\text{sam}}^\top \Phi \mathbf{p}_{j,\text{sam}} \mathbf{1}_{\{x \geq v_{j,\text{sam}}\}} &\xrightarrow{a.s.} \int_{-\infty}^x \delta(\lambda) dF(\lambda). \end{aligned}$$

At the same time, since p/n_1 and p/n both go to $c > 0$, Theorem 4.1 of Bai and Silverstein (2010) tells us that the two limits F_1 and F are equal, and since $\delta_1(\cdot)$ and $\delta(\cdot)$ depend on F_1 and F respectively (both depend on the same c also; see equation (2.7) and (2.9) in Lam (2016a) for more details), we must have $\delta_1(\cdot) = \delta(\cdot)$ also.

With the above, (5.16) immediately implies the results we need. Hence it remains to show (5.15).

To prove the first and third results of (5.15), consider for $k = 0, 1$,

$$\begin{aligned}
p^{-1} \text{tr}((\tilde{\Phi}_1 - z\mathbf{I}_p)^{-1} \Phi^k) &= p^{-1} \text{tr}((\mathbf{I}_p - (\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1}(\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1))^{-1}(\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1} \Phi^k) \\
&= p^{-1} \text{tr}((\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1} \Phi^k) + R, \quad \text{with} \\
R &= \sum_{j \geq 1} p^{-1} \text{tr}[(\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1}(\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1)]^j (\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1} \Phi^k,
\end{aligned}$$

where R comes from a Neumann series expansion. Such an expansion is valid since

$$\begin{aligned}
r &:= \|(\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1}(\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1)\| \leq \|(\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1}\| \cdot \|\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1\| \\
&\leq \frac{1}{|\text{Im}(z)|} \cdot \max_{\ell \in I_1} \left| \frac{1}{\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p} - 1 \right| \|\tilde{\Phi}_{1,\text{sam}}\| \\
&\leq \frac{\|\Phi^{1/2}\|^2}{|\text{Im}(z)|} \cdot \max_{\ell \in I_1} \left| \frac{1}{\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p} - 1 \right| \left\| n_1^{-1} \sum_{\ell \in I_1} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top \right\| \\
&\leq \frac{\|\Phi^{1/2}\|^2 (1 + \sqrt{c})^2}{|\text{Im}(z)|} \cdot \max_{\ell \in I_1} \left| \frac{1}{\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p} - 1 \right| \xrightarrow{a.s.} 0,
\end{aligned}$$

where we used Lemma S.2 of Lam (2016b) to conclude $\|(\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1}\| \leq 1/|\text{Im}(z)|$, and we used Theorem 5.11 of Bai and Silverstein (2010) to conclude that $\|n_1^{-1} \sum_{\ell \in I_1} \mathbf{Z}_\ell \mathbf{Z}_\ell^\top\| \leq (1 + \sqrt{c})^2$ almost surely, since p/n_1 , like p/n , goes to $c > 0$ also. Finally, the term $\max_{\ell \in I_1} \left| \frac{1}{\mathbf{Z}_\ell^\top \Phi \mathbf{Z}_\ell / p} - 1 \right| \xrightarrow{a.s.} 0$ by the last part of the proof of Lemma 1. With this,

$$|R| \leq \sum_{j \geq 1} r^j \|\Phi^k\| \cdot \|(\tilde{\Phi}_{1,\text{sam}} - z\mathbf{I}_p)^{-1}\| \leq \frac{r \|\Phi^k\|}{(1-r)|\text{Im}(z)|} \xrightarrow{a.s.} 0,$$

so that we have proved the first and third results in (5.15). For the other two results, the proof follows exactly the same lines as before after replacing $\tilde{\Phi}_{1,\text{sam}}$ by $\tilde{\Phi}_{\text{sam}}$ and $\tilde{\Phi}_1$ by $\tilde{\Phi}$. This completes the proof of the lemma. $\square$

Proof of Theorem 2. Define $\theta = \int_0^1 \gamma_t^2 dt$, $\hat{\theta} = \text{tr}(\Sigma_p^{\text{RCV}})/p$ and $\Phi_{\text{Ideal}} = \mathbf{P} \text{diag}(\mathbf{P}^\top \Phi \mathbf{P}) \mathbf{P}^\top$. We can easily see that

$$EL(\Sigma_p, \hat{\Sigma}_p) \leq 1 - \left(\frac{p^{-1/2} \|(\hat{\theta} - \theta) \hat{\Phi}\|_F}{p^{-1/2} \theta \|\Phi_{\text{Ideal}} - \Phi\|_F} + \frac{p^{-1/2} \|\hat{\Phi} - \Phi\|_F}{p^{-1/2} \|\Phi_{\text{Ideal}} - \Phi\|_F} \right)^{-2}, \quad (5.17)$$

where $p^{-1/2} \|(\hat{\theta} - \theta) \hat{\Phi}\|_F \leq |\hat{\theta} - \theta| \max_{1 \leq i \leq p} \mathbf{p}_{1i}^\top \tilde{\Phi}_2 \mathbf{p}_{1i} \xrightarrow{a.s.} 0$ by Lemma 1 and the proof of Theorem 1. Since we are equivalently assuming that $p^{-1/2} \|\Sigma_{\text{Ideal}} - \Phi\|_F$ is not going to 0, it remains to show that the rightmost term above is going to 1. To this end, observe that

$$\frac{p^{-1} \|\hat{\Phi} - \Phi\|_F^2}{p^{-1} \|\Phi_{\text{Ideal}} - \Phi\|_F^2} = \frac{p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^\top \tilde{\Phi}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^\top \Phi \mathbf{p}_{1i})^2}{p^{-1} \|\mathbf{P} \text{diag}(\mathbf{P}^\top \Phi \mathbf{P}) \mathbf{P}^\top - \Phi\|_F^2} + \frac{p^{-1} \|\mathbf{P}_1 \text{diag}(\mathbf{P}_1^\top \Phi \mathbf{P}_1) \mathbf{P}_1^\top - \Phi\|_F^2}{p^{-1} \|\mathbf{P} \text{diag}(\mathbf{P}^\top \Phi \mathbf{P}) \mathbf{P}^\top - \Phi\|_F^2}.$$

By Assumptions 4 and 5, and the results of Lemma 2, we have

$$\begin{aligned} p^{-1} \|\mathbf{P}_1 \text{diag}(\mathbf{P}_1^\top \boldsymbol{\Phi} \mathbf{P}_1) \mathbf{P}_1^\top - \boldsymbol{\Phi}\|_F^2 &= p^{-1} \text{tr}(\boldsymbol{\Phi}^2) - p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i})^2 \\ &\xrightarrow{a.s.} \int \tau^2 dH(\tau) - \int \delta_1^2(\lambda) dF_1(\lambda) \\ &= \int \tau^2 dH(\tau) - \int \delta^2(\lambda) dF_s(\lambda), \end{aligned}$$

which is non-zero if $\boldsymbol{\Phi} \neq \mathbf{I}_p$. This is also the almost sure limit of $p^{-1} \|\boldsymbol{\Phi}_{\text{Ideal}} - \boldsymbol{\Phi}\|_F^2$, and hence the rightmost term in (5.17) is indeed going to 1 if we can also show that $p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i})^2 \xrightarrow{a.s.} 0$. By Lemma 1,

$$p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i})^2 \leq \max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i}} \right| \cdot \max_{1 \leq i \leq p} \mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i} \xrightarrow{a.s.} 0.$$

This completes the proof for the Frobenius loss.

For the inverse Stein's loss, by Lemma 2, we have

$$\begin{aligned} p^{-1} L(\boldsymbol{\Sigma}_p, \boldsymbol{\Sigma}_{\text{Ideal}}) &= p^{-1} \sum_{i=1}^p \log(\mathbf{p}_i^\top \boldsymbol{\Phi} \mathbf{p}_i) - p^{-1} \sum_{i=1}^p \log(v_{n,i}) \\ &\xrightarrow{a.s.} \int \log(\delta(\lambda)) dF(\lambda) - \int \log(\tau) dH(\tau), \end{aligned}$$

where $v_{n,i}$ is the i th largest eigenvalue of $\boldsymbol{\Phi}$. Now consider the decomposition

$$p^{-1} L(\boldsymbol{\Sigma}_p, \hat{\boldsymbol{\Sigma}}_p) = I_1 + I_2 + I_3 + I_4 + I_5, \quad \text{where}$$

$$I_1 = \log(\hat{\theta}/\theta),$$

$$I_2 = \left(\frac{\theta}{\hat{\theta}} - 1 \right) p^{-1} \sum_{i=1}^p \left(\frac{\mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i}} \right),$$

$$I_3 = p^{-1} \sum_{i=1}^p \left(\frac{\mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i}} \right) - 1,$$

$$I_4 = p^{-1} \sum_{i=1}^p \log \left(\frac{\mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Phi}}_2 \mathbf{p}_{1i}} \right),$$

$$I_5 = p^{-1} \sum_{i=1}^p \log(\mathbf{p}_{1i}^\top \boldsymbol{\Phi} \mathbf{p}_{1i}) - p^{-1} \sum_{i=1}^p \log(v_{n,i}).$$

We can prove easily that I_1, I_2, I_3 and I_4 are all almost surely 0 by the proof of Theorem 1, and the result of Lemma 1. By Lemma 2, we can show that

$$I_5 \xrightarrow{a.s.} \int \log(\delta(\lambda)) dF(\lambda) - \int \log(\tau) dH(\tau),$$

so that $p^{-1}L(\Sigma_p, \Sigma_{\text{Ideal}})/p^{-1}L(\Sigma_p, \hat{\Sigma}_p) \xrightarrow{a.s.} 1$, showing $EL(\Sigma_p, \hat{\Sigma}_p) \xrightarrow{a.s.} 0$. This completes the proof of the theorem. $\square$

Proof of Theorem 3. In light of the results from Lemma 2, we can follow exactly the same lines as in the proof of Theorem 6 in Lam (2016a) to prove this theorem. $\square$

Proof of Theorem 4. We have

$$\begin{aligned} p^{1/2} \|\mathbf{w}_{\text{opt}}\|_{\infty} &\leq \frac{p^{1/2} \|\hat{\Sigma}_p^{-1}\|_1}{\mathbf{1}_p^{\top} \hat{\Sigma}_p^{-1} \mathbf{1}_p} = \frac{p^{1/2} \|\hat{\Phi}^{-1}\|_{\infty}}{\mathbf{1}_p^{\top} \hat{\Phi}^{-1} \mathbf{1}_p} \leq \frac{p^{1/2} \cdot p^{1/2} \lambda_{\min}^{-1}(\hat{\Phi})}{p \lambda_{\min}(\hat{\Phi}^{-1})} \\ &= \frac{\lambda_{\max}(\hat{\Phi})}{\lambda_{\min}(\hat{\Phi})} \leq \frac{\max_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \tilde{\Phi}_2 \mathbf{p}_{1i}}{\min_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \tilde{\Phi}_2 \mathbf{p}_{1i}} \xrightarrow{a.s.} \frac{\max_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \Phi \mathbf{p}_{1i}}{\min_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \Phi \mathbf{p}_{1i}} \\ &\leq \frac{\lambda_{\max}(\Phi)}{\lambda_{\min}(\Phi)} = \text{Cond}(\Phi), \end{aligned}$$

where we used the results of Lemma 1 for the almost sure convergence. For the actual risk bound,

$$\begin{aligned} pR(\hat{\Sigma}_p) &= \frac{p \mathbf{1}_p^{\top} \hat{\Sigma}_p^{-1} \Sigma_p \hat{\Sigma}_p^{-1} \mathbf{1}_p}{(\mathbf{1}_p^{\top} \hat{\Sigma}_p^{-1} \mathbf{1}_p)^2} \leq \frac{p \cdot p \lambda_{\max}^2(\hat{\Sigma}_p^{-1}) \lambda_{\max}(\Sigma_p)}{p^2 \lambda_{\min}^2(\hat{\Sigma}_p^{-1})} = \left(\frac{\lambda_{\max}(\hat{\Phi})}{\lambda_{\min}(\hat{\Phi})} \right)^2 \lambda_{\max}(\Sigma_p) \\ &= \left(\frac{\max_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \tilde{\Phi}_2 \mathbf{p}_{1i}}{\min_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \tilde{\Phi}_2 \mathbf{p}_{1i}} \right)^2 \lambda_{\max}(\Sigma_p) \xrightarrow{a.s.} \left(\frac{\max_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \Phi \mathbf{p}_{1i}}{\min_{1 \leq i \leq p} \mathbf{p}_{1i}^{\top} \Phi \mathbf{p}_{1i}} \right)^2 \lambda_{\max}(\Sigma_p) \\ &\leq \left(\frac{\lambda_{\max}(\Phi)}{\lambda_{\min}(\Phi)} \right)^2 \lambda_{\max}(\Sigma_p) = \text{Cond}^2(\Phi) \lambda_{\max}(\Sigma_p). \end{aligned}$$

This completes the proof of the theorem. $\square$

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